What Do the Collocations of Two Kings, 夫差 and 勾践, Reveal in the Wu Yue Chunqiu?

Abstract and Source

This report intends to show what the book Wu Yue Chunqiu 《吴越春秋》 reveals through a search of the collocations of two target words: 夫差, the king of Wu, and 勾践, the king of Yue. With the help of AI to clean and sort the data, I then calculated the statistical significance of the collocations. The results show that the characteristics of the two kings are different — or, rather, that the writer intends to describe them differently. The collocation results indicate that 夫差's collocates form a profile of campaigns, overlordship, succession and killing, whereas 勾践's collocates form a profile of submission, ministerial reliance, hesitation and reputation-management. Combined with the fact that the writer of the Wu Yue Chunqiu was born in 会稽山阴, the district mainly ruled by Yue, the words suggest that the writer had a partiality for Yue, praising Yue for its successful revenge while also criticizing the tragedy that befell its ministers.

The main source of this report is the Wu Yue Chunqiu (a .txt file, 145 KB), attributed to Zhao Ye of the Eastern Han. It is a literary reconstruction of Spring-and-Autumn-period Wu–Yue history, and it is a public-domain simplified-character edition taken from the Daizhige 殆知阁, a database of ancient Chinese texts. It is not a critical edition, because the text is a later, rhetorically shaped retelling rather than a contemporary record; its content mainly concerns how the story is told, not merely what happened. After cleaning, the corpus holds 2,470 sentences / 23,308 word tokens / 9,200 word types.

Processing and Method

First, the text of the Wu Yue Chunqiu was cleaned with an LLM agent. At the very beginning I deleted the running head and the ten 卷 (chapter titles), and then used jieba (a Chinese text segmenter) to segment the text, one line per sentence, using the most common segmentation without forcing lengths, each word separated by a space.

Because co-occurrence was measured with find_collocates from qhchina (a toolkit for Chinese-language analysis), I chose the horizon of my target words to be 5, meaning that I extracted the five words on each side of the target word, requiring at least two window co-occurrences. I then used a contingency table to calculate and sort the collocations by significance and strength. For each target word, a 2×2 contingency table is built relative to the target.

Here, using the p-value alone (a p-value is the probability of seeing a co-occurrence at least this strong if the two events were independent; if the value is below 0.05, the pair can be

called statistically significant) is not enough to check whether a collocation is strongly connected to my target word, because p measures reliability, not strength: it rises with both effect size and frequency. Strength is reported separately by logDice and mutual information (MI), which depend only on how tightly the two words are associated.

Results and Interpretation

In the outcome, 勾践 occurs 44 times in 43 sentences and 夫差 23 times in 21 sentences. 夫差 has 9 significant collocates — 波, 北伐, 太子, 被, 诸侯, 君王, 齐, 诛, 吴 — while 勾践 has 11 — 东海, 范蠡, 役, 下臣, 稽首, 闻于, 再拜, 臣, 不忍, 二, 越王. The two share no significant collocates, which means the descriptions of the two are strongly contrasted, at least in this book. In an early run, 卷 and 外传 were not cleaned completely and so became significant collocates; this was later corrected. At the beginning I also required each sentence to be six words long, but I changed it to segment sentences as they are commonly read and used, and the candidates changed in that run. After that I found that the collocations contained many function words such as 从 and 不, so I removed all the connectives and deleted the collocations whose raw p-value was above 0.05.

Some collocates show a contradictory condition, as with 勾践 + 役 (役臣) and 范蠡. 役 is by far the strongest partner of 勾践 (MI 6.38, ratio 83), yet its p-value (1.44×10^{-4}) is no smaller than that of 范蠡 (1.43×10^{-4}), whose MI is less than half (2.92). Reading the original text, 役 occurs only twice, both times in the phrase 东海役臣勾践 / 东海役臣臣孤勾践, which is why its MI is high; 范蠡 occurs 6 times throughout the whole story of Yue, for example 下臣勾践从小臣范蠡 / 乃使相国范蠡进曰：越王勾践窃有二遗女 / 范蠡从吴欲去，恐勾践未返. This means 范蠡 has less exclusivity than 役, but he is also significant with 勾践 because 范蠡 is a narrative collocation. 役 refers to 勾践's self-degraded status after his defeat by Wu, showing that this set of humiliating terms is a performative diplomatic register. 范蠡 points to the relationship between 勾践 and his ministers: he is a character rather than a term, appearing in various stories, so statistically he is not as closely associated with 勾践. This is one part of the contradictory ranking of significance and strength; other parts also show that this is precisely why significance and strength must be read together.

The distinct profiles emerging from the collocation data strongly support the hypothesis that the Wu Yue Chunqiu constructs sharply contrasting images for the two kings. For 夫差, the significant collocates such as 北伐, 诸侯 and 诛 form a clear semantic field of military aggression, hegemonic ambition and violent retribution. These words paint a portrait of a ruler defined by external conquest and a desire for dominance. Conversely, 勾践's collocations are different: words such as 役, 稽首 and 不忍 reveal a narrative of submission, humiliation and extreme patience, so 勾践's profile is diplomatic and endurance-oriented. This is also supported by the bamboo manuscript Yue gong qi shi 越公其事. Yuri Pines (Early China 45, 2022) argues that this text transmits 勾践's story as a didactic historical narrative centred on self-strengthening:

the king's recovery rests on his own example and conduct, not on impersonal law. Pines identifies 勾践's "five policies": promoting agriculture, securing trust, attracting immigrants, enlarging the army, and disciplining the people. That two early texts (Yue gong qi shi and the Wu Yue Chunqiu) both show a positive view of 勾践's endurance and revenge is precisely the character-centred framing that my collocation results detect.

Limitations

These results are suggestive, not decisive. For example, the window can be changed to any other number; if it is set to 3 or 10, the significant collocations might be different. Also, both target words I chose are low-frequency (44 and 23 tokens), so most collocates rest on only two or three observations, which may lead to unstable statistical values. Finally, and most importantly, collocations cannot show meaning, causation or intent. They show only that two words co-occur more than chance predicts.

Reference

Pines, Yuri. "Didactic Narrative and the Art of Self-Strengthening: Reading the Bamboo Manuscript Yue gong qi shi 越公其事." Early China 45 (2022): 375–412. https://doi.org/10.1017/eac.2022.2.